Muneeb Rehman
CT-25092

Lab 5

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```cpp
// Exercise 1: Demonstrating public, protected, and private inheritance

#include <iostream>

class Base
{
    private:
        int privateInt;

    protected:
        int protectedInt;

    public:
        int publicInt;

        int getPrivateInt()
        {
            return privateInt;
        }

        int getProtectedInt()
        {
            return protectedInt;
        }

        int getPublicInt()
        {
            return publicInt;
        }

        void setPrivateInt(int val)
        {
            privateInt = val;
        }

        void setProtectedInt(int val)
        {
            protectedInt = val;
        }

        void setPublicInt(int val)
        {
            publicInt = val;
        }
};

class publicChild : public Base
{
    public:
        void displayAccess()
        {
            std::cout << "\n--- Public Inheritance ---";

            std::cout << "\nPrivate Int (via getter): " << getPrivateInt();

            std::cout << "\nProtected Int (direct access): " << protectedInt;

            std::cout << "\nPublic Int (direct access): " << publicInt;
        }
};

class protectedChild : protected Base
{
```

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```cpp
    public:
        void displayAccess()
        {
            std::cout << "\n\n--- Protected Inheritance ---";

            std::cout << "\nPrivate Int (via getter): " << getPrivateInt();

            std::cout << "\nProtected Int (direct access): " << protectedInt;

            std::cout << "\nPublic Int (direct access): " << publicInt;
        }

        void setValues(int priv, int prot, int pub)
        {
            setPrivateInt(priv);
            setProtectedInt(prot);
            setPublicInt(pub);
        }
};

class privateChild : private Base
{
    public:
        void displayAccess()
        {
            std::cout << "\n\n--- Private Inheritance ---";

            std::cout << "\nPrivate Int (via getter): " << getPrivateInt();

            std::cout << "\nProtected Int (direct access): " << protectedInt;

            std::cout << "\nPublic Int (direct access): " << publicInt;
        }

        void setValues(int priv, int prot, int pub)
        {
            setPrivateInt(priv);
            setProtectedInt(prot);
            setPublicInt(pub);
        }
};

int main()
{
    publicChild obj1;
    obj1.setPrivateInt(10);
    obj1.setProtectedInt(20);
    obj1.setPublicInt(30);
    obj1.displayAccess();

    protectedChild obj2;
    obj2.setValues(40, 50, 60);
    obj2.displayAccess();

    privateChild obj3;
    obj3.setValues(70, 80, 90);
    obj3.displayAccess();

    std::cout << "\n\nSummary:";
    std::cout << "\n- Public Inheritance: Public and Protected members accessible, Private needs getter";
    std::cout << "\n- Protected Inheritance: All base members become protected in child";
    std::cout << "\n- Private Inheritance: All base members become private in child\n";
}
```

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```
--- Public Inheritance ---
Private Int (via getter): 10
Protected Int (direct access): 20
Public Int (direct access): 30

--- Protected Inheritance ---
Private Int (via getter): 40
Protected Int (direct access): 50
Public Int (direct access): 60

--- Private Inheritance ---
Private Int (via getter): 70
Protected Int (direct access): 80
Public Int (direct access): 90

Summary:
- Public Inheritance: Public and Protected members accessible, Private needs getter
- Protected Inheritance: All base members become protected in child
- Private Inheritance: All base members become private in child

--------------------------------
Process exited after 0.1742 seconds with return value 0
Press any key to continue . . .
```

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```cpp
// Exercise 2: Teacher class with derived classes for different subjects

#include <iostream>
#include <string.h>

class Teacher
{
    private:
        char* Name;
        int Age;
        char* Institute;

    public:
        Teacher()
        {
            Name = NULL;
            Age = 0;
            Institute = NULL;
        }

        ~Teacher()
        {
            if (Name != NULL)
            {
                delete[] Name;
            }
            if (Institute != NULL)
            {
                delete[] Institute;
            }
        }

        const char* getName()
        {
            return Name;
        }

        int getAge()
        {
            return Age;
        }

        const char* getInstitute()
        {
            return Institute;
        }

        void setName(const char* name)
        {
            if (Name != NULL)
            {
                delete[] Name;
            }
            Name = new char[strlen(name) + 1];
            strcpy(Name, name);
        }

        void setAge(int age)
        {
            Age = age;
        }

        void setInstitute(const char* inst)
```

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```cpp
        }

        void setInstitute(const char* inst)
        {
            if (Institute != NULL)
            {
                delete[] Institute;
            }
            Institute = new char[strlen(inst) + 1];
            strcpy(Institute, inst);
        }
};

class HumanitiesTeacher : public Teacher
{
    private:
        char Department[20];
        char* CourseName;
        char* Designation;

    public:
        HumanitiesTeacher()
        {
            strcpy(Department, "Humanities");
            CourseName = NULL;
            Designation = NULL;
        }

        ~HumanitiesTeacher()
        {
            if (CourseName != NULL)
            {
                delete[] CourseName;
            }
            if (Designation != NULL)
            {
                delete[] Designation;
            }
        }

        const char* getDepartment()
        {
            return Department;
        }

        const char* getCourseName()
        {
            return CourseName;
        }

        const char* getDesignation()
        {
            return Designation;
        }

        void setCourseName(const char* course)
        {
            if (CourseName != NULL)
            {
                delete[] CourseName;
            }
            CourseName = new char[strlen(course) + 1];
            strcpy(CourseName, course);
```

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```cpp
            }
            CourseName = new char[strlen(course) + 1];
            strcpy(CourseName, course);
        }

        void setDesignation(const char* desig)
        {
            if (Designation != NULL)
            {
                delete[] Designation;
            }
            Designation = new char[strlen(desig) + 1];
            strcpy(Designation, desig);
        }

        void display()
        {
            std::cout << "\n--- Humanities Teacher Details ---";
            std::cout << "\nName: " << getName();
            std::cout << "\nAge: " << getAge();
            std::cout << "\nInstitute: " << getInstitute();
            std::cout << "\nDepartment: " << Department;
            std::cout << "\nCourse Name: " << CourseName;
            std::cout << "\nDesignation: " << Designation;
            std::cout << "\n";
        }
};

class ScienceTeacher : public Teacher
{
    private:
        char Department[20];
        char* CourseName;
        char* Designation;

    public:
        ScienceTeacher()
        {
            strcpy(Department, "Science");
            CourseName = NULL;
            Designation = NULL;
        }

        ~ScienceTeacher()
        {
            if (CourseName != NULL)
            {
                delete[] CourseName;
            }
            if (Designation != NULL)
            {
                delete[] Designation;
            }
        }

        const char* getDepartment()
        {
            return Department;
        }

        const char* getCourseName()
        {
            return CourseName;
```

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```cpp
        const char* getCourseName()
        {
            return CourseName;
        }

        const char* getDesignation()
        {
            return Designation;
        }

        void setCourseName(const char* course)
        {
            if (CourseName != NULL)
            {
                delete[] CourseName;
            }
            CourseName = new char[strlen(course) + 1];
            strcpy(CourseName, course);
        }

        void setDesignation(const char* desig)
        {
            if (Designation != NULL)
            {
                delete[] Designation;
            }
            Designation = new char[strlen(desig) + 1];
            strcpy(Designation, desig);
        }


        void display()
        {
            std::cout << "\n--- Science Teacher Details ---";
            std::cout << "\nName: " << getName();
            std::cout << "\nAge: " << getAge();
            std::cout << "\nInstitute: " << getInstitute();
            std::cout << "\nDepartment: " << Department;
            std::cout << "\nCourse Name: " << CourseName;
            std::cout << "\nDesignation: " << Designation;
            std::cout << "\n";
        }
};

class MathsTeacher : public Teacher
{
    private:
        char Department[20];
        char* CourseName;
        char* Designation;

    public:
        MathsTeacher()
        {
            strcpy(Department, "Maths");
            CourseName = NULL;
            Designation = NULL;
        }

        ~MathsTeacher()
        {
            if (CourseName != NULL)
            {
```

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```cpp
        {
            if (CourseName != NULL)
            {
                delete[] CourseName;
            }
            if (Designation != NULL)
            {
                delete[] Designation;
            }
        }

        const char* getDepartment()
        {
            return Department;
        }

        const char* getCourseName()
        {
            return CourseName;
        }

        const char* getDesignation()
        {
            return Designation;
        }

        void setCourseName(const char* course)
        {
            if (CourseName != NULL)
            {
                delete[] CourseName;
            }
            CourseName = new char[strlen(course) + 1];
            strcpy(CourseName, course);
        }

        void setDesignation(const char* desig)
        {
            if (Designation != NULL)
            {
                delete[] Designation;
            }
            Designation = new char[strlen(desig) + 1];
            strcpy(Designation, desig);
        }

        void display()
        {
            std::cout << "\n--- Maths Teacher Details ---";
            std::cout << "\nName: " << getName();
            std::cout << "\nAge: " << getAge();
            std::cout << "\nInstitute: " << getInstitute();
            std::cout << "\nDepartment: " << Department;
            std::cout << "\nCourse Name: " << CourseName;
            std::cout << "\nDesignation: " << Designation;
            std::cout << "\n";
        }
};

int main()
{
    char tempInput[100];
    int ageInput;
```

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```cpp
{
    char tempInput[100];
    int ageInput;

    HumanitiesTeacher teacher1;
    std::cout << "=== Enter Humanities Teacher Details ===\n";
    std::cout << "Name: ";
    std::cin.getline(tempInput, 100);
    teacher1.setName(tempInput);

    std::cout << "Age: ";
    std::cin >> ageInput;
    teacher1.setAge(ageInput);
    std::cin.ignore();

    std::cout << "Institute: ";
    std::cin.getline(tempInput, 100);
    teacher1.setInstitute(tempInput);

    std::cout << "Course Name: ";
    std::cin.getline(tempInput, 100);
    teacher1.setCourseName(tempInput);

    std::cout << "Designation: ";
    std::cin.getline(tempInput, 100);
    teacher1.setDesignation(tempInput);

    ScienceTeacher teacher2;
    std::cout << "\n=== Enter Science Teacher Details ===\n";
    std::cout << "Name: ";
    std::cin.getline(tempInput, 100);
    teacher2.setName(tempInput);

    std::cout << "Age: ";
    std::cin >> ageInput;
    teacher2.setAge(ageInput);
    std::cin.ignore();

    std::cout << "Institute: ";
    std::cin.getline(tempInput, 100);
    teacher2.setInstitute(tempInput);

    std::cout << "Course Name: ";
    std::cin.getline(tempInput, 100);
    teacher2.setCourseName(tempInput);

    std::cout << "Designation: ";
    std::cin.getline(tempInput, 100);
    teacher2.setDesignation(tempInput);

    MathsTeacher teacher3;
    std::cout << "\n=== Enter Maths Teacher Details ===\n";
    std::cout << "Name: ";
    std::cin.getline(tempInput, 100);
    teacher3.setName(tempInput);

    std::cout << "Age: ";
    std::cin >> ageInput;
    teacher3.setAge(ageInput);
    std::cin.ignore();

    std::cout << "Institute: ";
    std::cin.getline(tempInput, 100);
```

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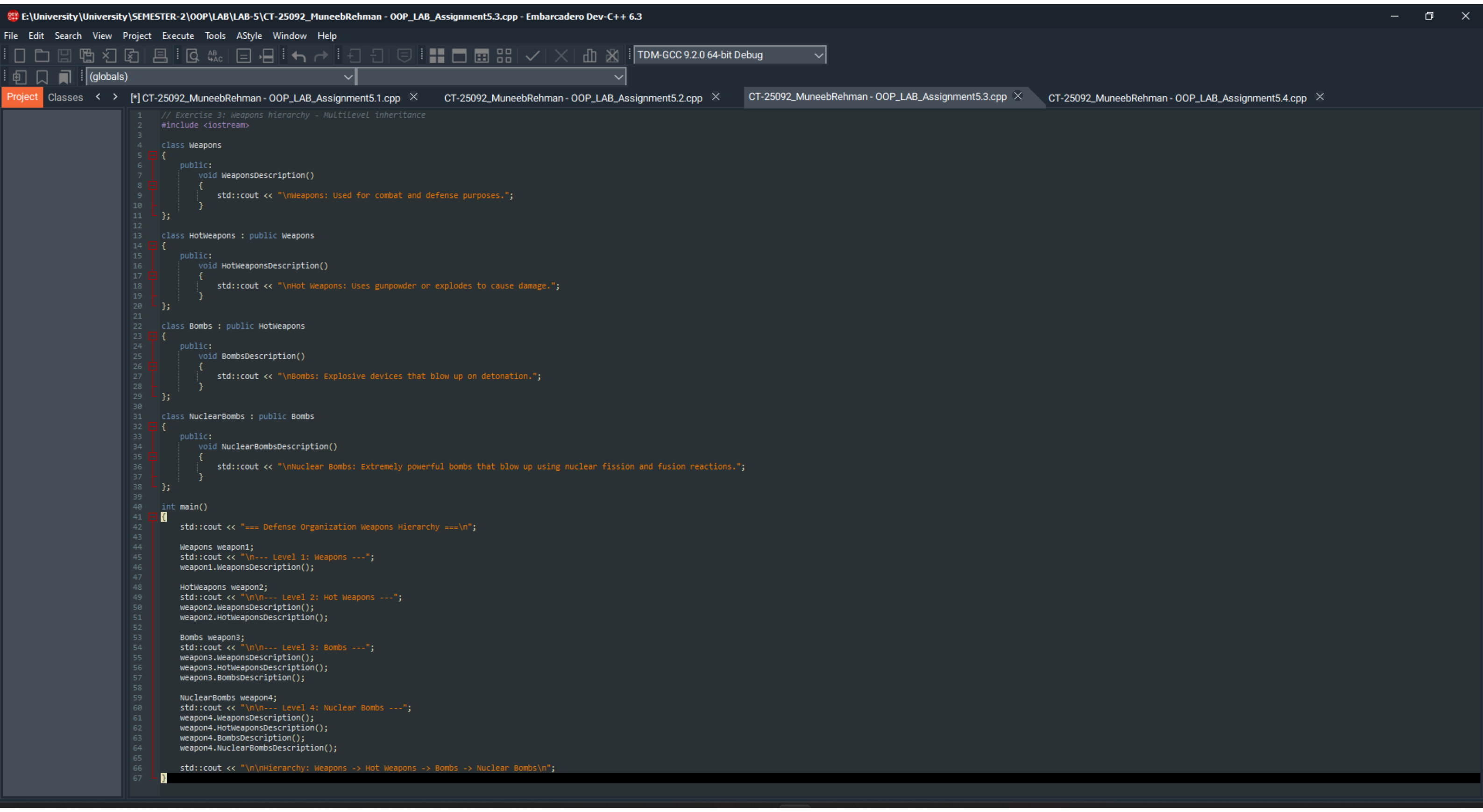
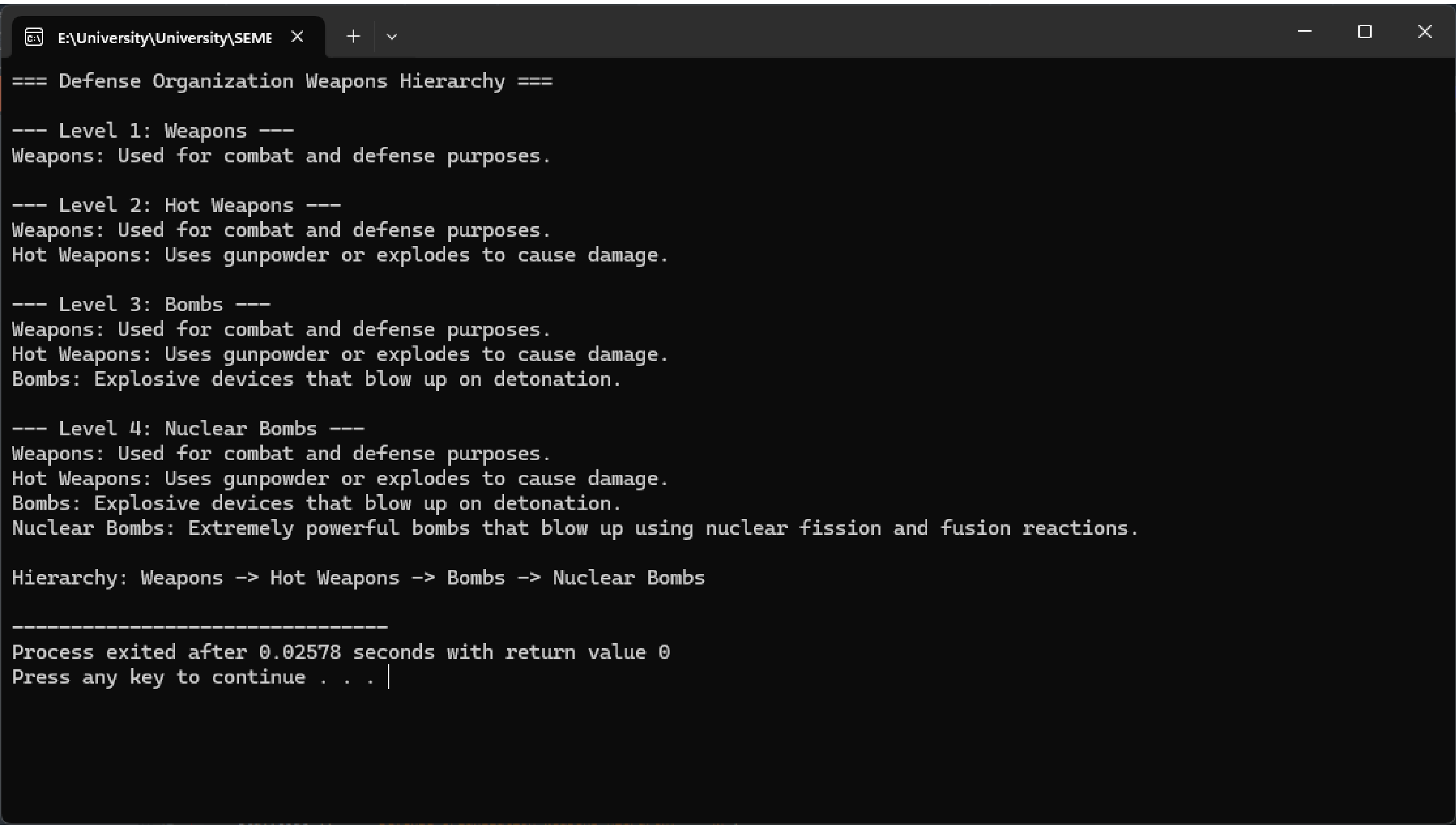
```cpp
    teacher1.setInstitute(tempInput);

    std::cout << "Course Name: ";
    std::cin.getline(tempInput, 100);
    teacher1.setCourseName(tempInput);

    std::cout << "Designation: ";
    std::cin.getline(tempInput, 100);
    teacher1.setDesignation(tempInput);

    ScienceTeacher teacher2;
    std::cout << "\n=== Enter Science Teacher Details ===\n";
    std::cout << "Name: ";
    std::cin.getline(tempInput, 100);
    teacher2.setName(tempInput);

    std::cout << "Age: ";
    std::cin >> ageInput;
    teacher2.setAge(ageInput);
    std::cin.ignore();

    std::cout << "Institute: ";
    std::cin.getline(tempInput, 100);
    teacher2.setInstitute(tempInput);

    std::cout << "Course Name: ";
    std::cin.getline(tempInput, 100);
    teacher2.setCourseName(tempInput);

    std::cout << "Designation: ";
    std::cin.getline(tempInput, 100);
    teacher2.setDesignation(tempInput);

    MathsTeacher teacher3;
    std::cout << "\n=== Enter Maths Teacher Details ===\n";
    std::cout << "Name: ";
    std::cin.getline(tempInput, 100);
    teacher3.setName(tempInput);

    std::cout << "Age: ";
    std::cin >> ageInput;
    teacher3.setAge(ageInput);
    std::cin.ignore();

    std::cout << "Institute: ";
    std::cin.getline(tempInput, 100);
    teacher3.setInstitute(tempInput);

    std::cout << "Course Name: ";
    std::cin.getline(tempInput, 100);
    teacher3.setCourseName(tempInput);

    std::cout << "Designation: ";
    std::cin.getline(tempInput, 100);
    teacher3.setDesignation(tempInput);

    std::cout << "\n\n=== Teacher Information ===";
    teacher1.display();
    teacher2.display();
    teacher3.display();

    return 0;
}
```

```
=== Enter Humanities Teacher Details ===
Name: John Doe
Age: 32
Institute: NED University
Course Name: Ethical Behaviour
Designation: Assistant Professor

=== Enter Science Teacher Details ===
Name: John Smith
Age: 29
Institute: NED University
Course Name: Physics
Designation: Lecturer

=== Enter Maths Teacher Details ===
Name: Jane Doe
Age: 30
Institute: NED University
Course Name: Mathematics
Designation: Lecturer


=== Teacher Information ===
--- Humanities Teacher Details ---
Name: John Doe
Age: 32
Institute: NED University
Department: Humanities
Course Name: Ethical Behaviour
Designation: Assistant Professor

--- Science Teacher Details ---
Name: John Smith
Age: 29
Institute: NED University
Department: Science
Course Name: Physics
Designation: Lecturer

--- Maths Teacher Details ---
Name: Jane Doe
Age: 30
Institute: NED University
Department: Maths
Course Name: Mathematics
Designation: Lecturer

--------------------------------
Process exited after 142.6 seconds with return value 0
Press any key to continue . . .
```



```cpp
// Exercise 3: Weapons hierarchy - Multilevel inheritance
#include <iostream>

class Weapons
{
    public:
        void WeaponsDescription()
        {
            std::cout << "\nWeapons: Used for combat and defense purposes.";
        }
};

class HotWeapons : public Weapons
{
    public:
        void HotWeaponsDescription()
        {
            std::cout << "\nHot Weapons: Uses gunpowder or explodes to cause damage.";
        }
};

class Bombs : public HotWeapons
{
    public:
        void BombsDescription()
        {
            std::cout << "\nBombs: Explosive devices that blow up on detonation.";
        }
};

class NuclearBombs : public Bombs
{
    public:
        void NuclearBombsDescription()
        {
            std::cout << "\nNuclear Bombs: Extremely powerful bombs that blow up using nuclear fission and fusion reactions.";
        }
};

int main()
{
    std::cout << "=== Defense Organization Weapons Hierarchy ===\n";

    Weapons weapon1;
    std::cout << "\n--- Level 1: Weapons ---";
    weapon1.WeaponsDescription();

    HotWeapons weapon2;
    std::cout << "\n\n--- Level 2: Hot Weapons ---";
    weapon2.WeaponsDescription();
    weapon2.HotWeaponsDescription();

    Bombs weapon3;
    std::cout << "\n\n--- Level 3: Bombs ---";
    weapon3.WeaponsDescription();
    weapon3.HotWeaponsDescription();
    weapon3.BombsDescription();

    NuclearBombs weapon4;
    std::cout << "\n\n--- Level 4: Nuclear Bombs ---";
    weapon4.WeaponsDescription();
    weapon4.HotWeaponsDescription();
    weapon4.BombsDescription();
    weapon4.NuclearBombsDescription();

    std::cout << "\n\nHierarchy: Weapons -> Hot Weapons -> Bombs -> Nuclear Bombs\n";
}
```



```
=== Defense Organization Weapons Hierarchy ===

--- Level 1: Weapons ---
Weapons: Used for combat and defense purposes.

--- Level 2: Hot Weapons ---
Weapons: Used for combat and defense purposes.
Hot Weapons: Uses gunpowder or explodes to cause damage.

--- Level 3: Bombs ---
Weapons: Used for combat and defense purposes.
Hot Weapons: Uses gunpowder or explodes to cause damage.
Bombs: Explosive devices that blow up on detonation.

--- Level 4: Nuclear Bombs ---
Weapons: Used for combat and defense purposes.
Hot Weapons: Uses gunpowder or explodes to cause damage.
Bombs: Explosive devices that blow up on detonation.
Nuclear Bombs: Extremely powerful bombs that blow up using nuclear fission and fusion reactions.

Hierarchy: Weapons -> Hot Weapons -> Bombs -> Nuclear Bombs

--------------------------------
Process exited after 0.02578 seconds with return value 0
Press any key to continue . . .
```

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```cpp
// Exercise 4: Bakery billing system with inheritance

#include <iostream>
#include <string.h>

class Item
{
    protected:
        char* Name;
        int Quantity;

    public:
        Item()
        {
            Name = NULL;
            Quantity = 0;
        }

        ~Item()
        {
            if (Name != NULL)
            {
                delete[] Name;
            }
        }

        void setName(const char* name)
        {
            if (Name != NULL)
            {
                delete[] Name;
            }
            Name = new char[strlen(name) + 1];
            strcpy(Name, name);
        }

        void setQuantity(int qty)
        {
            Quantity = qty;
        }

        const char* getName()
        {
            return Name;
        }

        int getQuantity()
        {
            return Quantity;
        }

        virtual float calculateTotal()
        {
            return 0.0;
        }
};

class BakedGoods : public Item
{
    protected:
        float Discount;

    public:
```

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```cpp
        float Discount;

    public:
        BakedGoods()
        {
            Discount = 0.10;
        }

        float getDiscount()
        {
            return Discount;
        }
};

class Cakes : public BakedGoods
{
    private:
        float Price;

    public:
        Cakes()
        {
            Price = 600.0;
            setName("Cake");
        }

        float getPrice()
        {
            return Price;
        }

        float calculateTotal()
        {
            float total = Price * Quantity;
            float discountAmount = total * Discount;
            return total - discountAmount;
        }

        void displayBill()
        {
            std::cout << "\nItem: " << Name;
            std::cout << "\nQuantity: " << Quantity;
            std::cout << "\nPrice per unit: " << Price;
            std::cout << "\nDiscount: " << (Discount * 100) << "%";
            std::cout << "\nTotal: " << calculateTotal() << " PKR";
        }
};

class Bread : public BakedGoods
{
    private:
        float Price;

    public:
        Bread()
        {
            Price = 200.0;
            setName("Bread");
        }

        float getPrice()
        {
            return Price;
        }
```

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```cpp
        float getPrice()
        {
            return Price;
        }

        float calculateTotal()
        {
            float total = Price * Quantity;
            float discountAmount = total * Discount;
            return total - discountAmount;
        }

        void displayBill()
        {
            std::cout << "\nItem: " << Name;
            std::cout << "\nQuantity: " << Quantity;
            std::cout << "\nPrice per unit: " << Price;
            std::cout << "\nDiscount: " << (Discount * 100) << "%";
            std::cout << "\nTotal: " << calculateTotal() << " PKR";
        }
};

class Drinks : public Item
{
    private:
        float Discount;
        float Price;

    public:
        Drinks()
        {
            Discount = 0.05;
            Price = 100.0;
            setName("Drink");
        }

        float getDiscount()
        {
            return Discount;
        }

        float getPrice()
        {
            return Price;
        }

        float calculateTotal()
        {
            float total = Price * Quantity;
            float discountAmount = total * Discount;
            return total - discountAmount;
        }

        void displayBill()
        {
            std::cout << "\nItem: " << Name;
            std::cout << "\nQuantity: " << Quantity;
            std::cout << "\nPrice per unit: " << Price;
            std::cout << "\nDiscount: " << (Discount * 100) << "%";
            std::cout << "\nTotal: " << calculateTotal() << " PKR";
        }
};
```

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```cpp
        {
            float total = Price * Quantity;
            float discountAmount = total * Discount;
            return total - discountAmount;
        }

        void displayBill()
        {
            std::cout << "\nItem: " << Name;
            std::cout << "\nQuantity: " << Quantity;
            std::cout << "\nPrice per unit: " << Price;
            std::cout << "\nDiscount: " << (Discount * 100) << "%";
            std::cout << "\nTotal: " << calculateTotal() << " PKR";
        }
};

int main()
{
    int numCakes, numBread, numDrinks;
    float grandTotal = 0.0;

    std::cout << "=== Bakery Billing System ===\n";

    std::cout << "\nEnter quantity of Cakes: ";
    std::cin >> numCakes;
    Cakes cakeOrder;
    cakeOrder.setQuantity(numCakes);

    std::cout << "Enter quantity of Bread: ";
    std::cin >> numBread;
    Bread breadOrder;
    breadOrder.setQuantity(numBread);

    std::cout << "Enter quantity of Drinks: ";
    std::cin >> numDrinks;
    Drinks drinkOrder;
    drinkOrder.setQuantity(numDrinks);

    std::cout << "\n\n=== Customer Bill ===";

    if (numCakes > 0)
    {
        std::cout << "\n";
        cakeOrder.displayBill();
        grandTotal = grandTotal + cakeOrder.calculateTotal();
    }

    if (numBread > 0)
    {
        std::cout << "\n";
        breadOrder.displayBill();
        grandTotal = grandTotal + breadOrder.calculateTotal();
    }

    if (numDrinks > 0)
    {
        std::cout << "\n";
        drinkOrder.displayBill();
        grandTotal = grandTotal + drinkOrder.calculateTotal();
    }

    std::cout << "\n\n--- Grand Total: " << grandTotal << " PKR ---\n";
}
```

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```
=== Bakery Billing System ===

Enter quantity of Cakes: 24
Enter quantity of Bread: 32
Enter quantity of Drinks: 80


=== Customer Bill ===

Item: Cake
Quantity: 24
Price per unit: 600
Discount: 10%
Total: 12960 PKR

Item: Bread
Quantity: 32
Price per unit: 200
Discount: 10%
Total: 5760 PKR

Item: Drink
Quantity: 80
Price per unit: 100
Discount: 5%
Total: 7600 PKR

--- Grand Total: 26320 PKR ---

--------------------------------
Process exited after 17.74 seconds with return value 0
Press any key to continue . . .
```